

NOUPOORT, South Africa

WMO: 686330

Lat: 31.1864S

Lon: 24.9603E

Elev: 1495

StdP: 84.61

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.3	-0.9	-15.2	1.2	11.7	-13.2	1.4	13.9	10.5	11.6	9.7	10.9	0.9	130	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	32.6	14.4	31.1	14.2	29.7	14.1	18.7	24.0	18.1	23.5	17.5	23.0	5.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	14.8	20.6	16.5	14.1	20.2	15.8	13.5	19.8	60.1	23.6	57.7	23.5	55.7	23.0	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	9.4	8.4	-5.3	35.9	1.5	3.5	-6.4	38.4	-7.3	40.5	-8.2	42.4	-9.3	45.0
			-7.1	20.0	1.2	1.5	-7.9	21.1	-8.6	22.0	-9.3	22.8	-10.1	23.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	21.8	21.6	19.3	15.3	11.8	8.1	10.3	13.6	17.0	18.8	21.2
		DBStd	5.99	3.20	2.66	3.18	3.52	3.31	2.69	3.13	3.63	4.00	3.99	2.96
		HDD10.0	227	0	0	1	4	20	67	76	43	12	3	0
		HDD18.3	1493	8	5	27	101	204	306	322	251	147	73	41
		CDD10.0	2247	367	326	287	164	75	11	13	51	122	220	264
		CDD18.3	472	117	97	56	10	1	0	0	7	32	55	98
		CDH23.3	4166	1033	714	399	74	4	0	5	103	331	551	950
		CDH26.7	1344	383	225	96	11	0	0	0	21	92	172	344
(14)	Wind	WSAvg	3.7	4.3	3.9	3.6	3.1	2.8	3.0	3.1	3.3	3.6	4.3	4.6
(15)	Precipitation	PrecAvg	334	52	55	53	32	16	16	10	12	11	23	26
		PrecMax	508	122	167	110	92	60	43	39	77	65	108	74
		PrecMin	203	2	14	23	3	1	2	0	0	2	0	0
		PrecStd	81	33	34	21	21	17	12	10	16	17	23	18
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	33.4	31.5	27.7	23.8	19.1	19.6	23.9	29.6	32.0	32.8
			MCWB	15.2	14.7	15.4	13.8	11.5	9.2	7.9	9.4	11.7	12.9	13.4
		2%	DB	32.4	31.2	29.2	25.3	21.7	17.3	17.7	21.4	26.3	29.3	30.5
			MCWB	14.9	14.8	14.9	13.7	10.6	8.2	7.8	9.0	10.4	12.3	13.3
		5%	DB	30.6	29.4	27.4	23.5	19.9	15.8	16.1	19.3	23.9	27.0	28.7
			MCWB	14.9	15.1	14.8	12.9	10.2	7.5	7.4	8.3	9.9	11.9	13.0
		10%	DB	28.7	27.6	25.6	21.5	17.9	14.1	14.3	17.2	21.7	24.8	26.7
			MCWB	14.8	15.4	15.0	12.7	9.9	6.9	7.0	7.6	9.3	11.3	12.7
		0.4%	WB	19.2	19.7	19.4	17.2	13.7	11.1	10.2	12.5	14.2	15.7	17.7
			MCDB	24.2	25.3	23.6	22.3	17.8	16.2	15.2	18.1	20.3	24.6	22.0
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.3	19.0	18.2	15.9	12.5	9.7	9.1	11.1	12.9	14.7	16.6
			MCDB	23.5	24.3	23.4	20.4	17.8	14.3	15.0	16.6	21.2	23.2	22.2
		2%	WB	17.7	18.2	17.5	14.9	11.7	8.7	8.2	10.0	11.8	13.8	15.6
			MCDB	23.3	23.7	22.8	19.5	16.9	13.5	14.2	16.9	19.5	22.0	22.1
		5%	WB	17.1	17.6	16.8	14.1	10.8	7.9	7.4	8.9	10.7	13.0	14.7
			MCDB	23.1	23.0	22.2	18.9	16.1	12.6	13.3	15.6	19.6	21.7	22.1
		10%	WB	16.8	17.1	16.4	13.8	10.5	7.6	7.1	8.6	10.4	12.7	14.4
			MCDB	22.8	22.7	21.9	18.6	15.8	12.3	13.0	15.3	19.3	21.4	21.9
		0.4%	WB	19.2	19.7	19.4	17.2	13.7	11.1	10.2	12.5	14.2	15.7	17.7
			MCDB	24.2	25.3	23.6	22.3	17.8	16.2	15.2	18.1	20.3	24.6	22.0
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.5	11.9	12.0	11.5	11.9	12.4	12.7	13.1	14.3	13.8	13.5
			MCDBR	15.1	14.5	14.4	14.0	14.1	14.4	14.5	15.3	17.1	16.1	15.9
		5% WB	MCWBR	3.8	3.9	4.5	5.8	6.3	7.4	7.2	7.6	5.7	4.5	4.2
			MCWBR	11.1	10.7	10.9	11.2	11.3	12.0	12.7	12.2	13.5	13.0	12.0
				3.9	3.9	4.4	5.8	5.9	6.8	7.0	6.2	6.8	5.0	4.5
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.317	0.306	0.282	0.263	0.250	0.249	0.287	0.313	0.319	0.297	0.300
		taud	2.564	2.542	2.551	2.610	2.647	2.651	2.643	2.491	2.396	2.417	2.550	2.586
		Ebn at Noon	1029	1004	983	959	934	926	942	937	958	990	1036	1043
		Edh at Noon	107	106	99	83	71	67	70	92	113	119	108	106
(44)	All-Sky Solar Radiation	RadAvg	7.74	7.05	6.01	4.67	3.73	3.29	3.60	4.49	5.77	6.93	7.84	8.29
(45)		RadStd	0.42	0.41	0.29	0.20	0.16	0.18	0.17	0.25	0.30	0.26	0.37	0.42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page